

# 1 Generic transport equation

Some definitions:

$$\tilde{Q} = -G_{\parallel} A y' + G_{\parallel} B y + G_{\parallel} R \quad (1)$$

$$\tilde{Q}_{\rho} := \tilde{Q} - \dot{b} \rho y N \quad (2)$$

$$\kappa = \frac{\rho}{\rho_b} \quad (3)$$

$$\frac{\partial}{\partial t} \frac{y}{\rho} - \rho \dot{b} \frac{\partial y}{\partial \rho} = D(\Psi) + S \quad (4)$$

The ASTRA generic transport equation reads:

$$\begin{aligned} S y + P &= \frac{1}{G} \frac{\partial}{\partial t} (G \kappa y)_{\rho} + \frac{1}{V} \frac{\partial}{\partial \rho} (V M \tilde{Q}_{\rho}) = \frac{1}{G} \frac{\partial}{\partial t} (G \kappa y)_{\rho} + \frac{1}{V} \frac{\partial}{\partial \rho} (V M \tilde{Q}) - \frac{\dot{b}}{W} \frac{\partial}{\partial \rho} (\rho V M N y) = \\ &= \frac{1}{G} \frac{\partial}{\partial t} (G \kappa y)_x - x (\dot{r} - \dot{b}) \frac{1}{G} \frac{\partial}{\partial x} (G \kappa y) + \frac{1}{V} \frac{\partial}{\partial \rho} (V M \tilde{Q}) - \frac{\dot{b}}{W} \frac{\partial}{\partial \rho} (\rho V M N y) \end{aligned}$$

Substituting the definitions above, the explicit form reads:

$$\begin{aligned} &\frac{1}{G} \frac{\partial}{\partial t} (G \kappa y)_x + \frac{1}{V} \frac{\partial}{\partial \rho} \left[ V M \left( -G_{\parallel} A y' + G_{\parallel} B y + G_{\parallel} R \right) \right] = \\ &= S y + P + \frac{\dot{b}}{V} \frac{V}{W} \frac{\partial}{\partial \rho} (V M N \rho y) + (\dot{r} - \dot{b}) \frac{\rho}{V} \frac{V}{G} \frac{\partial}{\partial \rho} (G \kappa y) \end{aligned}$$

In implicit form, the transport equation looks like:

$$\begin{aligned} &\frac{1}{G} \frac{\partial}{\partial t} (G \kappa y)_x + \frac{1}{V} \frac{\partial}{\partial \rho} \left[ V M \left( -G_{\parallel} A y' + G_{\parallel} B y + G_{\parallel} R - \dot{b} \frac{V}{W} N \rho y - (\dot{r} - \dot{b}) \rho \frac{\kappa y}{M} \right) \right] = \\ &= S y + P - (\dot{r} - \dot{b}) \frac{G H y}{V} \frac{\partial}{\partial \rho} R \left( \frac{\rho V}{G} \right) - \dot{b} M N \rho y \frac{\partial}{\partial \rho} \left( \frac{V}{W} \right) \end{aligned}$$

# 2 Physical meaning of the argument variables

|                 | G    | H     | M        | $G_{\parallel}$     | N                      | V     | A             | B           | S   | $P$              | W        | R  |
|-----------------|------|-------|----------|---------------------|------------------------|-------|---------------|-------------|-----|------------------|----------|----|
| $\Psi$          | 1    | 1     | 1/G      | 1                   | $1/\rho$               | $C_1$ | $G_2$         | 0           | 0   | $j_{bs} + \dots$ | $1/\rho$ | 0  |
| n               | $V'$ | 1     | $1/V'$   | $G_{\parallel}$     | $V'$                   | $V'$  | Diff          | $V_{pinch}$ | SNN | SN               | $V'$     | 0  |
| $F_j$           | $V'$ | 1     | $1/V'$   | $G_{\parallel}$     | $V'$                   | $V'$  | Diff          | $V_p$       | SFN | SF               | $V'$     | 0  |
| $T_e$           | v53  | $n_e$ | $625/V'$ | $G_{\parallel}/625$ | $3/2 \ n_e \ V'^{5/3}$ | $V'$  | $\chi_e$      | $V_{pe}$    | PET | PE               | $V'$     | 0  |
| $U_{\parallel}$ | $V'$ | UPS   | 1        | 1                   | 1                      | $V'$  | $\chi_{\phi}$ | $V_{pu}$    | 0   | Torque           | $V'$     | Re |